

reduce costs and complexity. Also, Cascading Style Sheets (CSS) and JavaScript can be combined with HTML5 to provide a high-end UI experience and validation support.

How HTML5 mobile application development works

HTML5 is a mark-up language used to structure and display content on websites. It provides 'write-once-run-anywhere' methodology, which makes distribution much easier than the native apps. It contains a set of Web pages optimised for mobile devices and to deliver high-end video and audio data manipulation capabilities, graphics handling and offline support. It also adds elements like form controls, multimedia components and other APIs that support geo location services and drag-and-drop operations. Almost all popular browsers support HTML5, which enables developers to implement this language across different platforms and devices.

Although HTML is supported by almost all browsers, a slight difference in the mobile devices and browsers may affect how an application works and its look and feel. To overcome this, apps are developed at the level of the lowest possible denominator so that they can work effortlessly on high-end versions. Now let's look at how the development of HTML5 powered mobile applications takes place.

Generally, the development takes place with the help of various frameworks available in the market, each with its own pros and cons, which we will discuss later. These frameworks contain various libraries of CSS and JavaScript files that app developers can use for rapid development. Some frameworks also have the capability to handle most of the issues arising out of the apps on their own. HTML is based on the Document Object Model (DOM), which you can think of as a tree-like logical structure used to access and manipulate a document or Web page. Sometimes, navigating through a complex and lengthy DOM can slow the application; so these frameworks can help programmers to improve the performance by recycling and reusing different DOM components. The use of a framework also makes it effortless to implement various functionalities such as screen transitions or list scrolling, which depend on JavaScript. As a result, it becomes easier for beginners also to develop apps that can provide a high-end user experience (UX).

Hybrid apps: The future

Although HTML5 mobile app development is catching on fast and has come a long way, it still cannot compete with some of the capabilities provided by native apps, like multi-



Figure 1: The same HTML5 code will work on all platforms

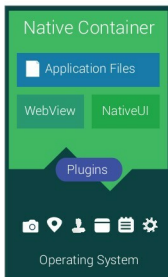


Figure 2: Architecture of hybrid apps

touch gestures and smooth integration with built-in mobile components, such as GPS, cameras, accelerometers and contacts. Offline storage and security are the other limitations of HTML5 apps, which also lose out to native apps on performance because the latter are usually packaged with machine code.

To get the best of both worlds, we can combine both HTML5 and native apps to build hybrid apps. A simple hybrid app is built using HTML5, JavaScript and CSS, but is wrapped inside a thin native container known as Webview (which you can think of as a windowless browser that's configured to run full screen). This enables mobile applications to access a device's hardware capabilities like the camera, etc, through JavaScript APIs like PhoneGap. Hybrid apps can also support offline operations and multi-touch gestures.

Though they look and feel like native apps due to being wrapped inside containers, hybrid apps still lag behind native apps a little bit in terms of performance due to the extra layers of abstraction involved in building them. You already know that native apps are installed on the device from the OS' app store

and HTML5 apps are viewed through Web browsers. But it may interest you to know how hybrid apps are installed or used. There are actually two ways -- local and server -- to implement hybrid apps in the device.

- **Local:** Just like a native app, one can package the source code containing HTML and JavaScript files inside the mobile binary.
- **Server:** Alternatively, one can also use a server to implement the mobile application.

If hybrid apps are well written, it is extremely difficult to determine how they have been developed. Since these apps combine the best of two worlds, they are definitely the future in mobility and can help organisations reduce the costs and liabilities associated with developing native apps.

Frameworks for hybrid app development

Having looked at HTML5, native and hybrid apps and their pros and cons, let's take a look at the top eight frameworks that you can use to develop your own hybrid mobile applications. The following top eight have been selected based on my own experience and usage.

1. **ionic:** If you are a beginner in the world of hybrid mobile application development, then the Ionic framework is the way to go. It is an open source framework (protected under the MIT licence) with a focus on performance and gets regularly updated, keeping it ahead of its competitors. It can support iOS, Android and Windows platforms. One of